

Expt No: 01

Date:

VERIFICATION OF GATES

Aim:-

To study and verify the truth table of logic gates.

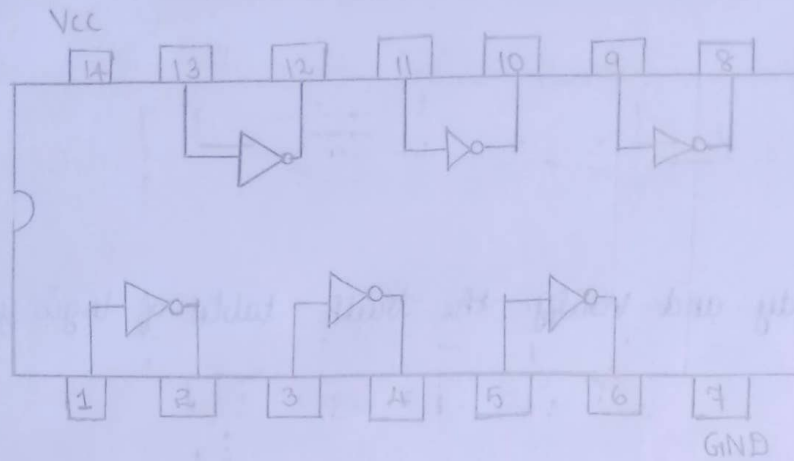
Apparatus Required:-

All the basic gates mentioned in the figure.

Procedure:

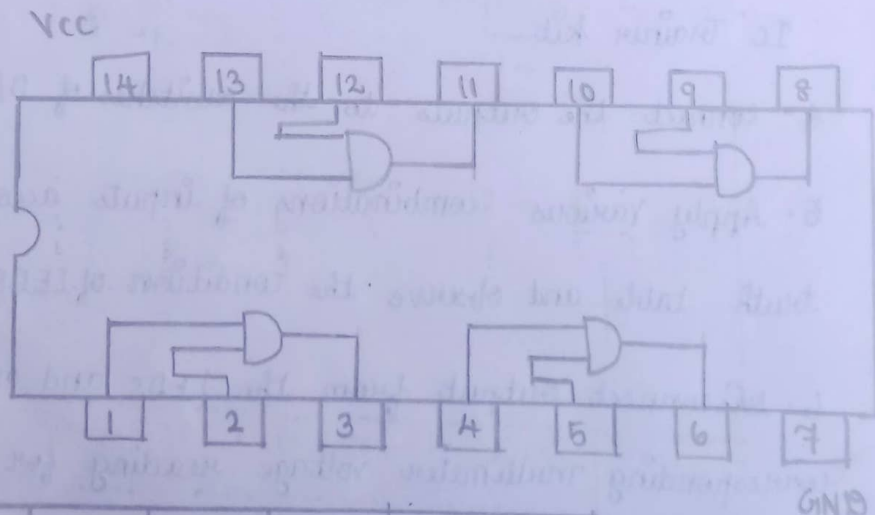
1. Place the IC on the IC Trainer kit
2. Connect VCC and ground to respective pins to the IC trainer kit
3. Connect the inputs to the input switches provided in the IC Trainer kit
4. Connect the outputs to the switches of O/P LEDs
5. Apply various combinations of inputs according to the truth table and observe the condition of LEDs.
6. Disconnect output from the LEDs and note down the corresponding multimeter voltage reading for various combinations of inputs.

Inverter Gate (NOT Gate) - 7404LS



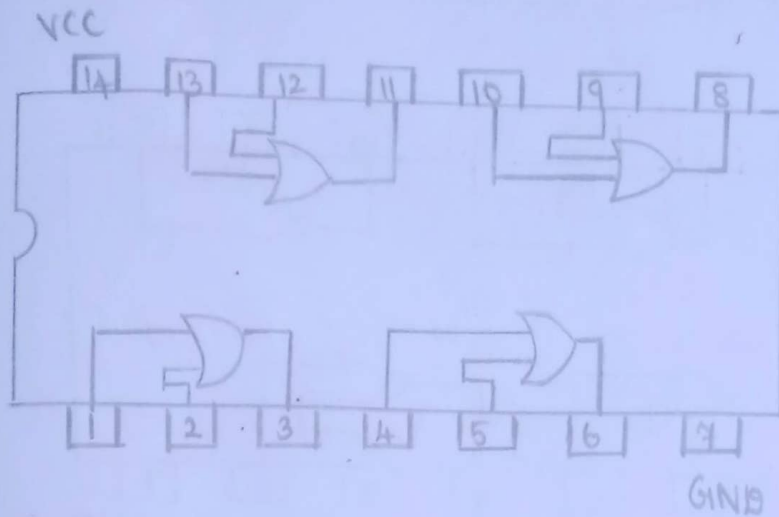
		Y_1	Y_2	Y_3	Y_4	Y_5	Y_6
A	O/P	(V)	(V)	(V)	(V)	(V)	(V)
0	1	5V	5V	5V	5V	5V	5V
1	0	0V	0V	0V	0V	0V	0V

2-Input AND Gate - 7408LS



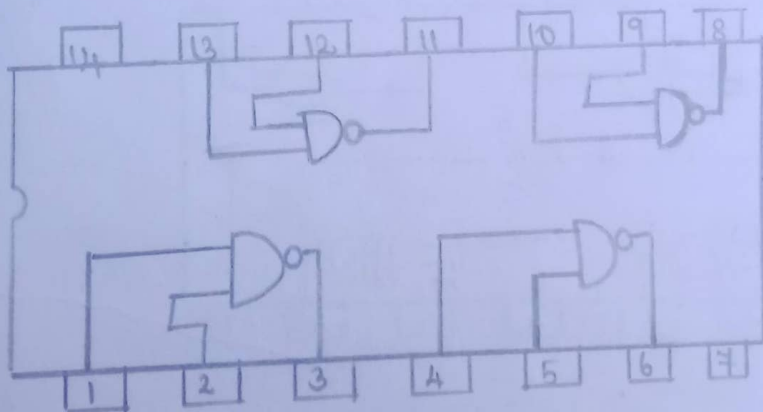
A	B	O/P	Y_1	Y_2	Y_3	Y_4
			(V)	(V)	(V)	(V)
0	0	0	0V	0V	0V	0V
0	1	0	0V	0V	0V	0V
1	0	0	0V	0V	0V	0V
1	1	1	5V	5V	5V	5V

Input OR Gate - 7432LS



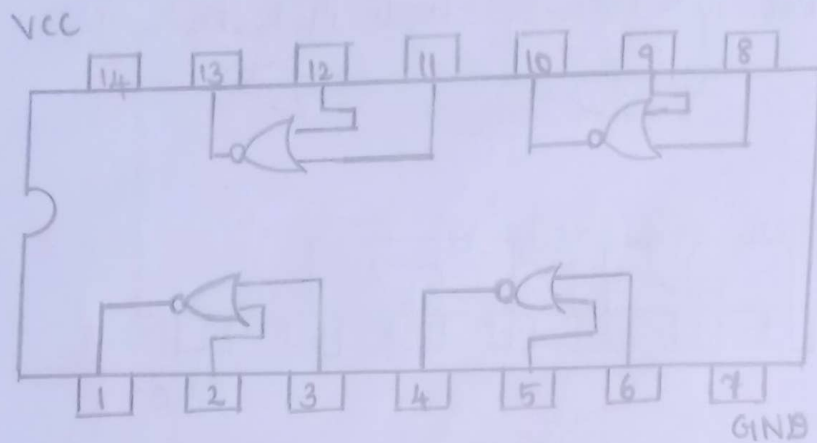
A	B	O/P	Y ₁ (V)	Y ₂ (V)	Y ₃ (V)	Y ₄ (V)
0	0	0	0V	0V	0V	0V
0	1	1	5V	5V	5V	5V
1	0	1	5V	5V	5V	5V
1	1	1	5V	5V	5V	5V

Input NAND Gate - 7400LS



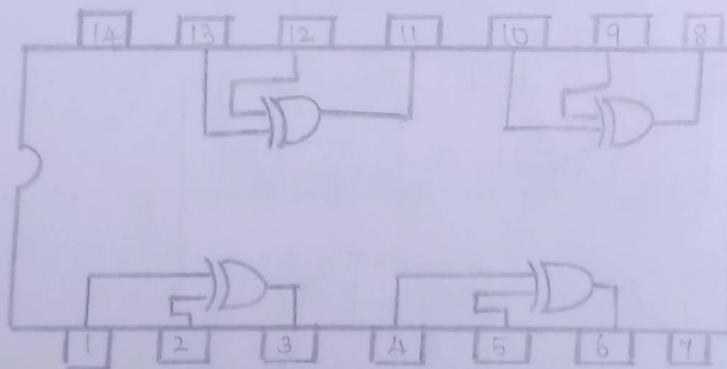
A	B	O/P	Y ₁ (V)	Y ₂ (V)	Y ₃ (V)	Y ₄ (V)
0	0	1	5V	5V	5V	5V
0	1	1	5V	5V	5V	5V
1	0	1	5V	5V	5V	5V
1	1	0	0V	0V	0V	0V

Input NOR Gate - 7402LS



A	B	O/P	Y ₁ (V)	Y ₂ (V)	Y ₃ (V)	Y ₄ (V)
0	0	1	5V	5V	5V	5V
0	1	0	0V	0V	0V	0V
1	0	0	0V	0V	0V	0V
1	1	0	0V	0V	0V	0V

Input EX-OR Gate 7486LS



A	B	O/P	Y ₁ (V)	Y ₂ (V)	Y ₃ (V)	Y ₄ (V)
0	0	0	0V	0V	0V	0V
0	1	1	5V	5V	5V	5V
1	0	1	5V	5V	5V	5V
1	1	0	0V	0V	0V	0V

CONCLUSION:-

Thus, the AND, OR, NOT, NAND, NOR and XOR gates were successfully verified using respective ICs.